

APPROVED PILOT GRANT · INTERNAL TEACHING AWARD

Cell Organization in Frontal-Bone Formation

AWARD \$50,000

pilot · one cycle

THE AIM

Measure whether cranial neural crest cells organize non-randomly as they build the frontal bone. We segment nuclei with Cellpose, map their spacing with a Voronoi diagram, and bind every claim to the most direct tool actually used. The frontal bone is chosen because it is the cleanest, most measurable case of these cells organizing.

WHY IT MATTERS

This is the same family of cells that, elsewhere in the face, builds the lip and palate that did not fuse in cases like Baby Mateo's. Keep the two separate: a cleft is a failure of facial parts to fuse, not a frontal-bone fate decision. We study the clean case here so the foundation underneath the harder ones is honest and measurable.

THE APPROACH (THREE ACTS)

Act I, Develop: spatial order in the building of the frontal bone. Act II, Differentiate: the bone-versus-cartilage decision, read through YAP, TAZ, and Sox9. Act III, Derail: the very same toolkit hijacked out of place in cancer. One toolkit, three settings.

THE TEAM

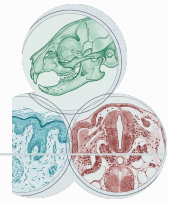
Suneeti Madhavan, Ph.D., lab lead, mentors and directs the project team. Piper Moore (Class of 2027) and the undergraduate research team run the imaging, segmentation, and analysis, and carry the discovery work that this pilot funds.

GROUNDING IN OUR LAB NOTES

The science is grounded in our own working notes, not invented for the game: the craniofacial research track and parent records in Notion; the durable research substrate in Open Brain; and the Cellpose, Voronoi, and ECM report plus the cleft-lip-and-palate research library in the Obsidian (OpenBrain) vault.

THE HONESTY CLAUSE

The award buys tools, not certainty. The top force and tension rungs stay out of reach for reasons drawn from the real lab: the in-the-body force instrument is not in this browser lab, and direct tissue stiffness needs dead tissue while the embryo is alive. You may claim only as high as your most direct tool reaches.



STUDENT BUDGET SHEET · MAKE YOUR DECISIONS BEFORE YOU CLICK

You do not get to buy everything. Every tool costs money, people, or both, and two rungs are not for sale at any price. Plan your spend, then commit.

THE HIRES (people you pay once)

Role (in the app)	Cost	What they unlock
Instrument Operator · "Sam"	\$12,000	Runs the big machines (time-lapse migration, the force rigs).
Data Interpreter · "Priya"	\$8,000	Turns a machine's raw signal into a result you are allowed to trust.

THE MEASUREMENT LADDER (what each tool buys)

Rung / tool	Measures	Cost	Needs	Claim it buys
1 Spatial layout	Where cells sit (Cellpose + Voronoi)	\$0	free in browser	Non-random spacing
2 ECM road (FN1)	The road they crawl on	\$8,000	none	Organized along FN1
3 Migration	How they move over time	\$30,000	Operator	Directed movement
4 Molecular force (FRET)	Pull between molecules	LOCKED	FRET rig (not in lab) + Operator + Interpreter	Not for sale here
5 Tissue stiffness (AFM)	Real tissue force	LOCKED	Dead tissue (embryo is alive) + Operator + Interpreter	Not for sale here

THREE WAYS TO SPEND IT (worked examples)

Plan	Buys	Spent	Grant left	Strongest honest claim
Free & honest	Spatial only	\$0	\$50,000	Cells are non-randomly spaced
Add the road	Spatial + ECM	\$8,000	\$42,000	Spacing tracks the FN1 road
Go for movement	Spatial + ECM + Migration + Operator	\$50,000	\$0	The cells move in a directed way

YOUR PLAN

Tools I will buy: _____	Hires: _____
Total spent: \$ _____	Grant left: \$ _____ Claim I can defend: _____

A locked rung is not a budget problem. More money cannot buy it in this lab. When you conclude, name what you did NOT measure.